ZAMBEZIA MARKET ASSESSMENT (MOPEIA, MORRUMBALA)

Market Functionality Index

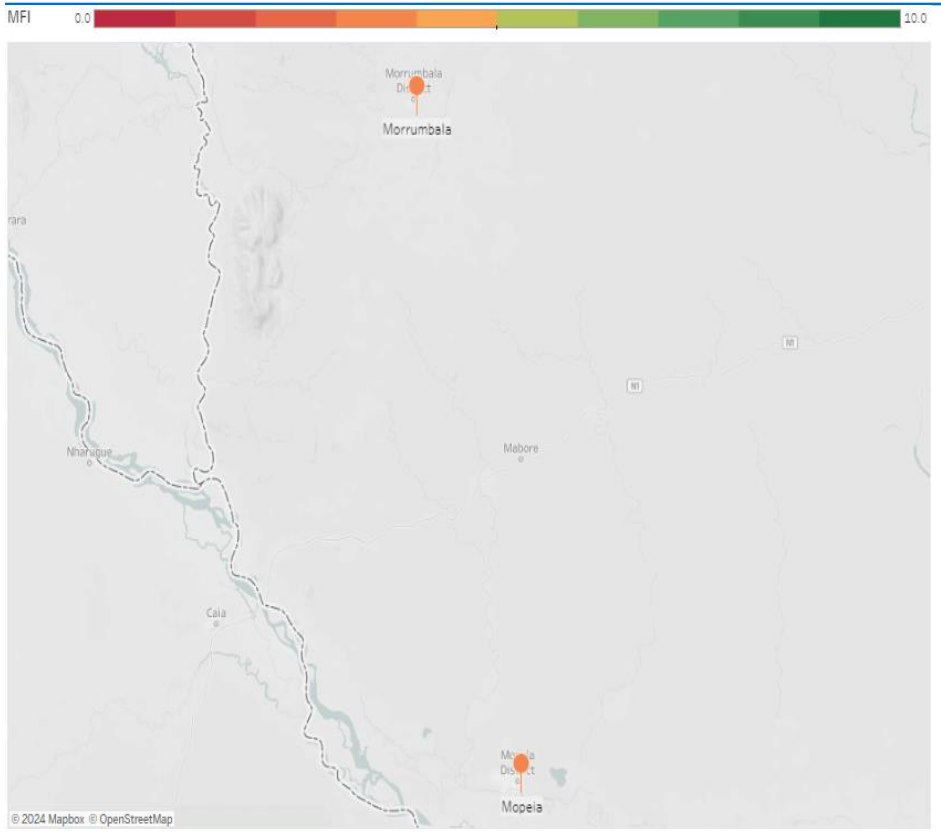
MARCH 2024



wfp.org

vam
food security analysis

ZAMBEZIA MARKET ASSESSMENT : OVERVIEW



OVERVIEW

Markets have become mainstream instruments for poverty reduction, and increased food & nutrition security. However, market dynamics, failures and shortcomings often diminish the desired impacts and/or long-term effects. In the assessed markets in Zambezia, the overall MFI score is 3.9, challenging the use of cash-based transfers, as there are very high levels of risk associated with the modality. While the assortment and availability dimensions, as per the standard MFI scoring do not highlight particular issues, in most of the assessed markets it was found that only 8% of the shops sell maize, 34% maize meal and 23% pulses, nuts and seeds.

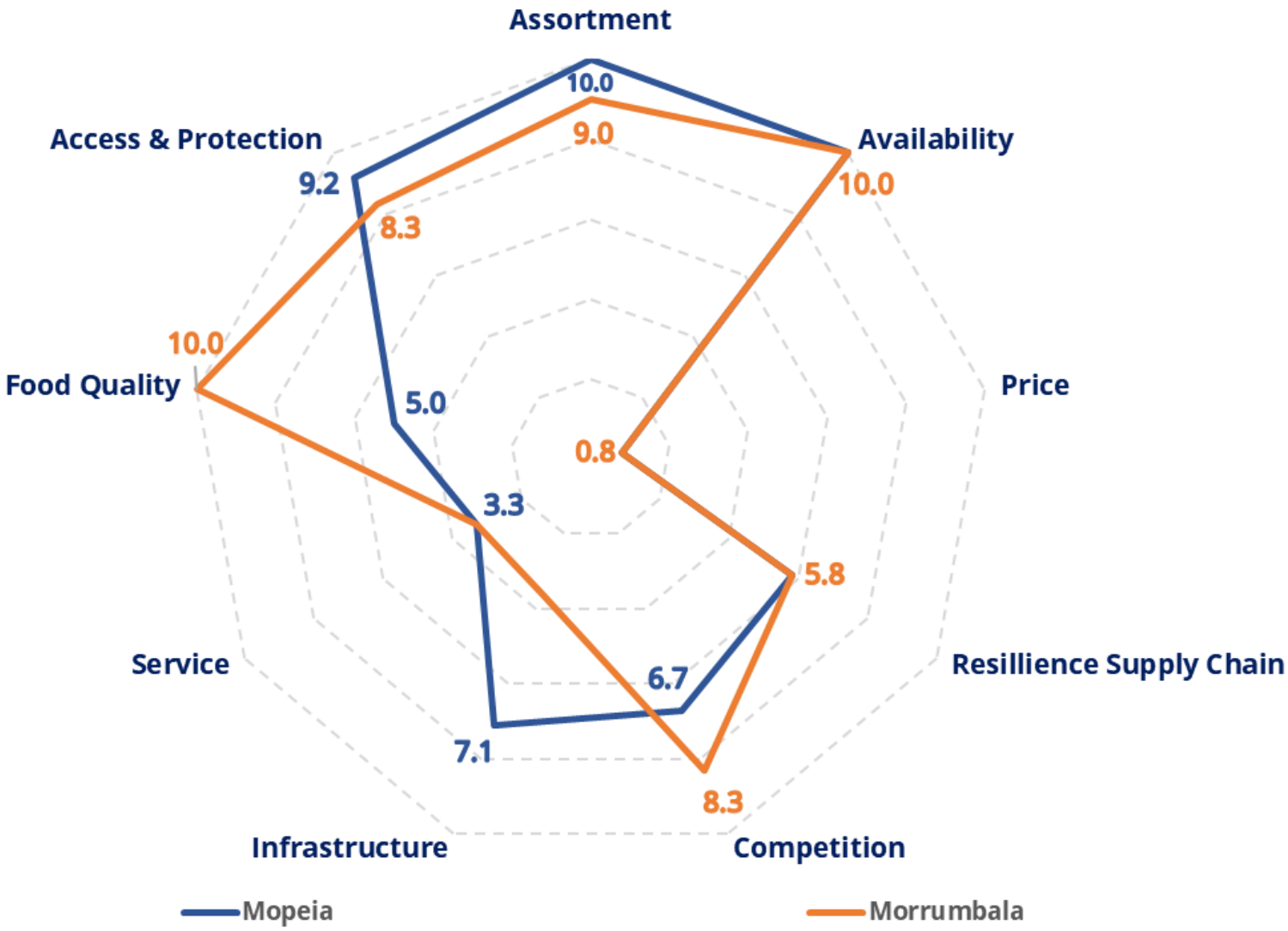
Price is the weakest dimension across the province, followed by the Service dimension, scoring lower than the average; hence several mitigation measures should be put in place to safeguard against price-related risks for cash-based transfers - these includes continuing to strengthen market monitoring, and adapting to price fluctuations, such as by calculating the minimum expenditure basket, and adjusting the transfer values as needed.

Moreover, many services expected in market premises are missing. including shelters close to the markets to host traders and other market players, water sources, toilets and sewage/waste collection systems. In some markets, issues with Quality of Food and Competition were also identified and must be addressed or mitigated before implementing CBT programs.

Market Name	Risk level for Market-based intervention	Assortment	Availability	Price	Resillience Supply Chain	Competition	Infrastructure	Service	Food Quality	Access & Protection	MFI
Mopeia	Very High Risk	10.0	10.0	0.8	5.8	6.7	7.1	3.3	5.0	9.2	3.9
Morrumbala	Very High Risk	9.0	10.0	0.8	5.8	8.3	3.6	3.3	10.0	8.3	3.9
Provincial Average		**9.5**	**10.0**	**0.8**	**5.8**	**7.5**	**5.4**	**3.3**	**7.5**	**8.8**	**3.9**

EMERGENCY PREPARATION MARKET ASSESSMENT - ZAMBEZIA : OVERVIEW

Figure 1.



FINDINGS BY MFI DIMENSION : ASSORTMENT

ASSORTMENT

The assortment of essential goods answers to two questions:

1. Can beneficiaries find all their essential needs in the market?

2. Do they have a wide range of choices?

Essential needs refer to what households require to ensure minimum living standards (items such as food, shelter, basic household items, safe drinking water, sanitation and hygiene or healthcare). To evaluate choice, the MFI investigates the maximum number of commodities or Stock Keeping Units (SKUs) sold in outlets in the market.

KEY FINDINGS

The average score for assortment was 9.5, which indicates that assessed markets offer a broad assortment of commodities. The assortment dimension in these two 2 districts shows good breadth in almost all categories for food and non-food items.

The essential products most available In the markets targeted by this assessment are:

- Food: Rice, Cassava, Sorghum, Vegetable Oil, Maize meal, Fish & Eggs,
- Non-food items: soap, school supplies, clothes, shoes and some construction materials.

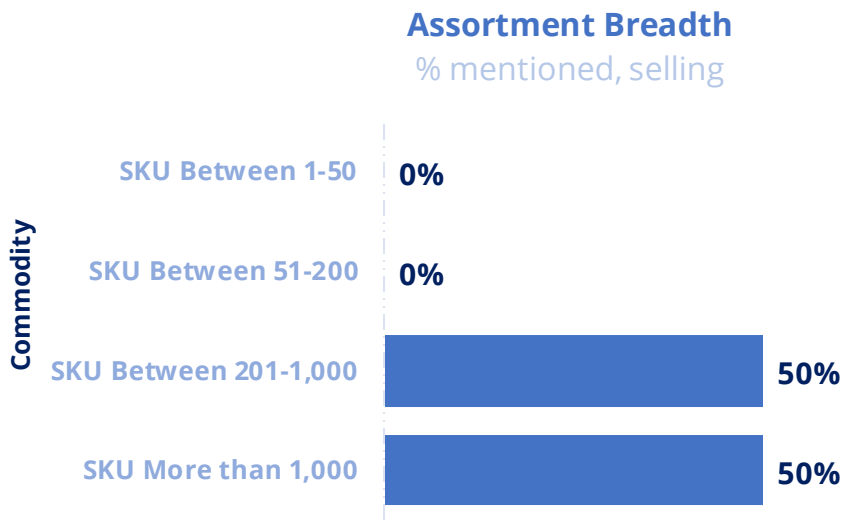
No shortage of any specific product was reported in those markets (i.e., there is a normal trading flow at the markets), but some issues were identified as only few (8%) retailers selling cereal food reported selling maize – considered a basic and important staple food in people's everyday lives and for WFP operations.

SCORE BY MARKET



FINDINGS BY MFI DIMENSION : ASSORTMENT (cont'd)

Table 1.



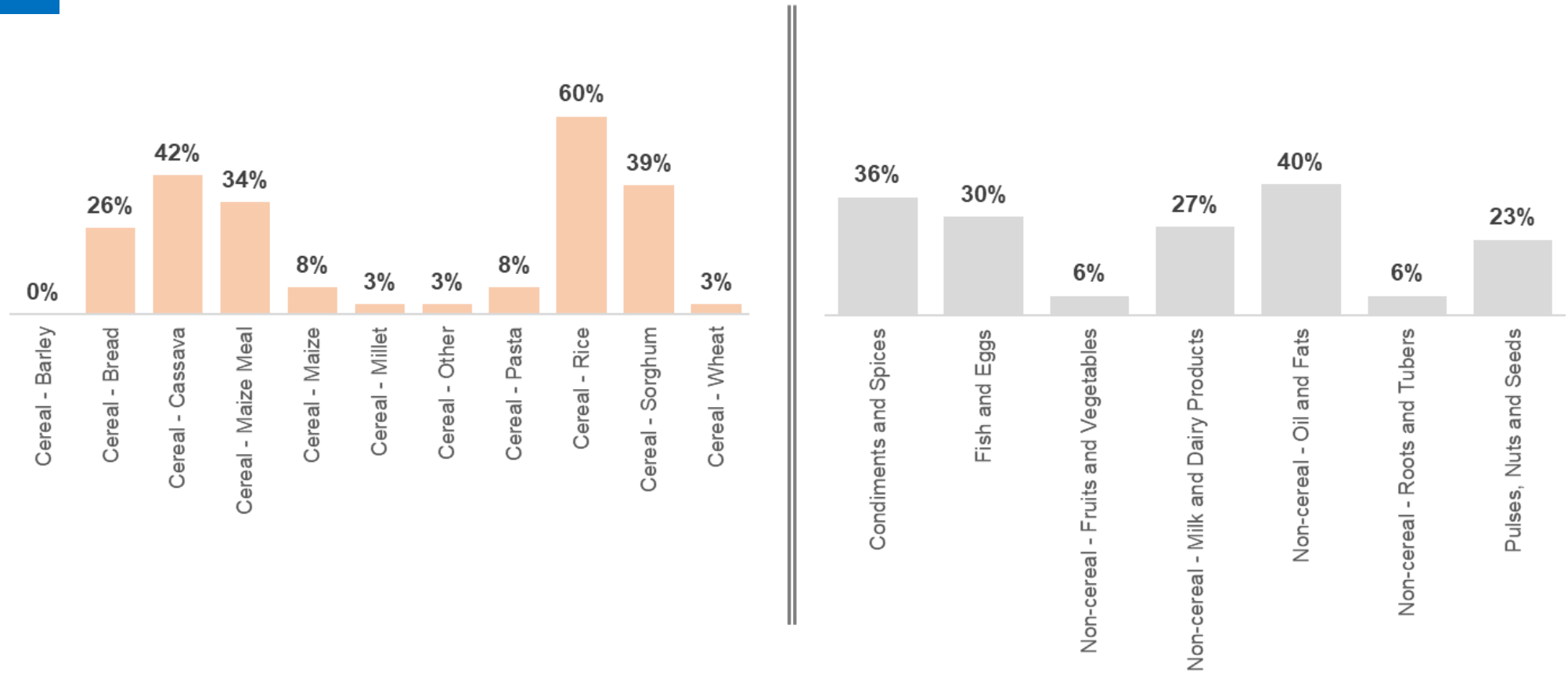
Commodity	Assortment Breadth % mentioned, selling	
	Mopeia	Morrumbala
Cereal Food	100%	100%
Non-Cereal Food	100%	100%
NFI - Communication	100%	100%
NFI - Education	100%	100%
NFI - Household Items	100%	100%
NFI - Health	100%	0%
NFI - Shelter	100%	100%
NFI - WASH	100%	100%

Overall, there was a wide breadth of distinct products available across markets. Traders reported a high product breadth of over 200 distinct items – also known as the number of stock-keeping units (SKUs) – nevertheless Morrumbala reported lack of assortment for Health NFIs.

FINDINGS BY MFI DIMENSION: ASSORTMENT

The proportion of traders selling each major food commodity is provided below:

Figure 2.



FINDINGS BY MFI DIMENSION : AVAILABILITY

AVAILABILITY

Availability is the physical presence of goods in sufficient quantities, thus adequate supplies in a functioning market. Availability answers to two questions:

1. Are there any key products that are scarce?
2. Are there any key products which are already running out?

Availability assesses whether certain products are scarce or likely to become scarcer in the short run.

KEY FINDINGS

THE AVERAGE REGIONAL SCORE FOR AVAILABILITY WAS 10.0, reflecting general physical presence of goods in sufficient quantities in retailers reporting selling those items. While a small percentage of retailers reported selling certain food commodities, such as maize or maize flour, they reported having them in sufficient quantities, few traders reported issues about running out of stock for some commodities, which means certain products are not scarce or likely to become scarcer in the short run.

❖ Scarcity

i) Cereals: only 16% of the traders selling cereals reported scarcity of cereals, from which 9% reported scarcity on rice.

ii) Food other than cereals: 34% of the traders selling other non-cereal foods reported scarcity of food other than cereals (Fruits & Vegetable and Legumes & Nuts)

iii) Non-food items: on average only 11% of traders reported scarcity of non-food items (mainly communication – 13%)

❖ Running out of commodities

i) Cereals: 18% of cereal traders feared running out of cereals, the most reported was Rice with 13%.

ii) Food other than cereals: 27% of the traders will run out of food other than cereals in a week from the surveyed day.

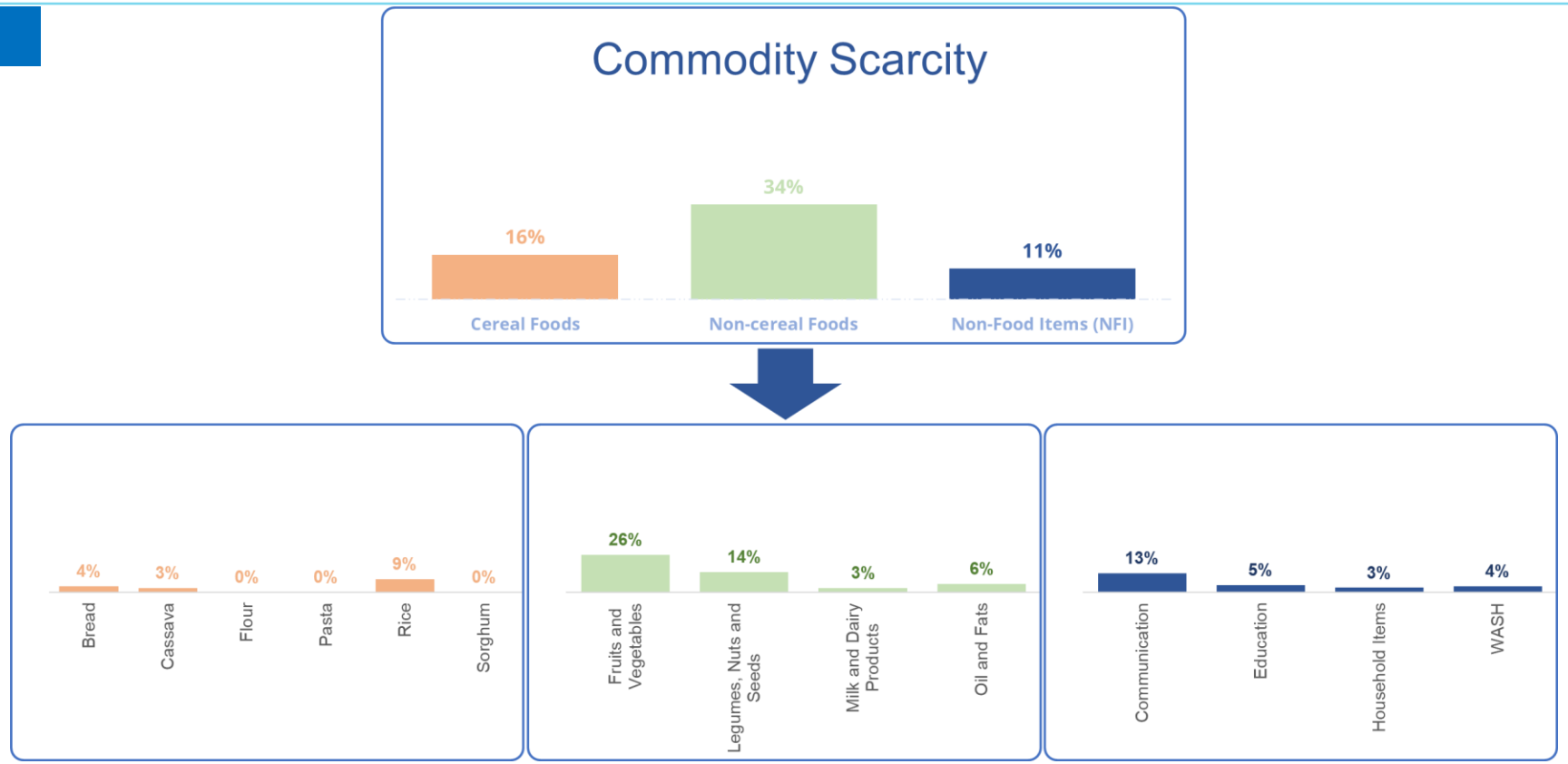
iii) NFIs: only 10% of the traders reported very small quantity of NFI,in particular on WASH items.

SCORE BY MARKET



FINDINGS BY MFI DIMENSION : AVAILABILITY (scarcity)

Figure 3.

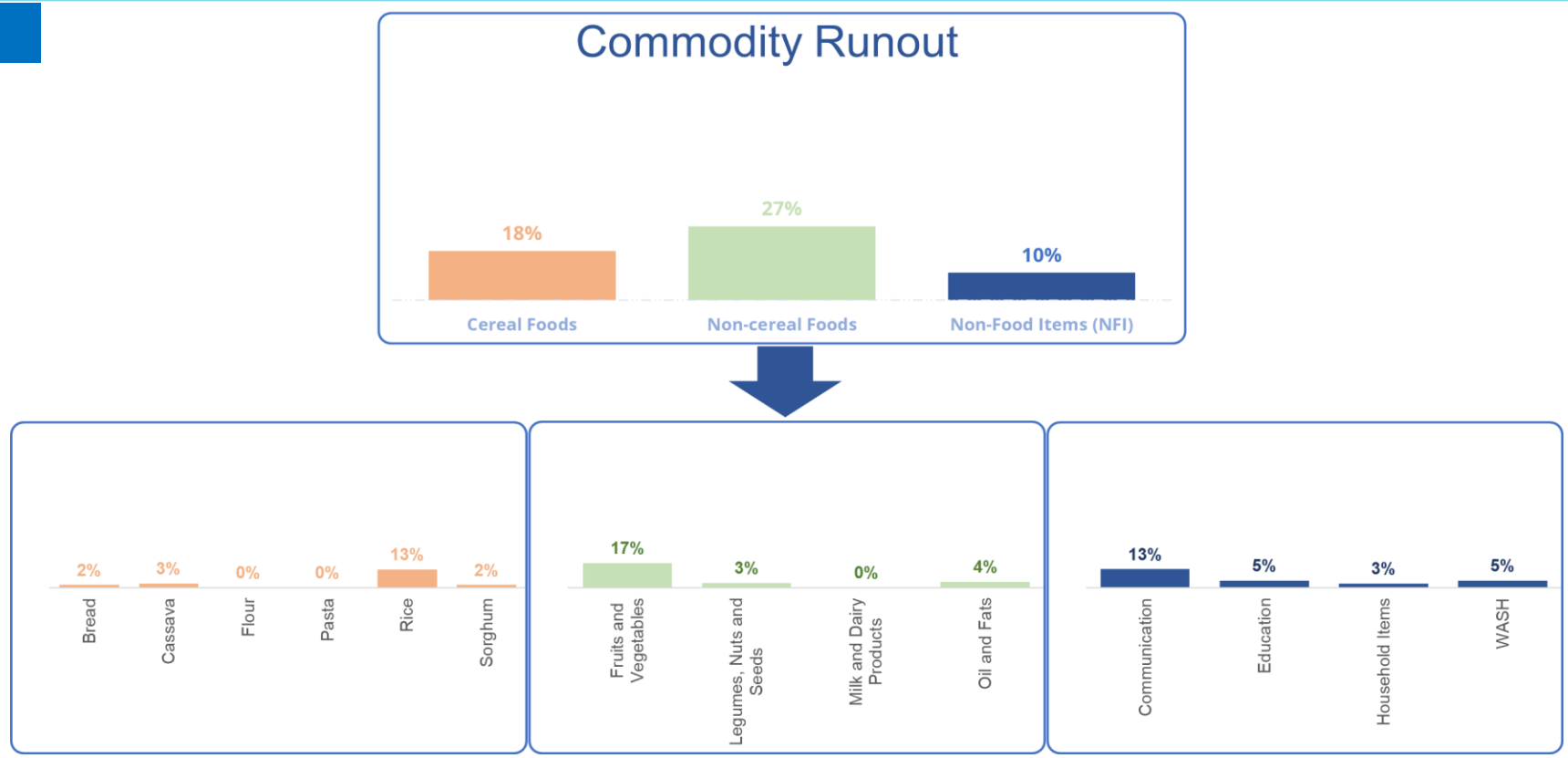


The markets assessed in Zambezia have a relatively adequate level of product availability, although traders reported scarcity (shortages) of some commodities within the three (3) categories: Cereals Foods (16%), Non-cereal Foods (34%) and Non-Food Items (11%). In the cereals category, only three products were reported, with rice being the most reported at just 9%, followed by bread at 4% and cassava at 3%.

When looking at non-cereal foods, 26% of retailers reported a scarcity of Fruit and Vegetables, while 14% reported on Legumes, Nuts and Seeds.

FINDINGS BY MFI DIMENSION : AVAILABILITY (runout)

Figure 4.



The availability pillar refers not only to the physical presence of goods in the market in sufficient quantities but it also assesses whether certain products are likely to get scarcer in the short run. No particular issues with goods availability were noticed across assessed markets in Zambezia, a small concern was noticed on a few items as 18% of retailers are running out of cereals, 27% of non-cereals and 10% on Non-Food Items.

Cereals: Rice was reported to be the most scarce cereal as 13% of retailers reported running out of quantity.

Non-cereal: 17% of retailers reported running out of Fruit and Vegetables

Non-food : Communication Items were reported the most scarce items considering that 13% of retailers reported running out of quantity.

FINDINGS BY MFI DIMENSION : PRICE

PRICE

Affordable, stable, predictable prices are key characteristics of functioning markets. The price dimensions answers two questions:

1. Were there significant price increases in the past month?
2. Are prices predictable or are they subject to unpredictable variations?

The price dimension assess stability of prices and the nature of any price changes. The critical aspect is unpredictability rather than variability itself. In markets with high seasonal changes, volatility can be largely predictable and, thus, not necessarily harmful because market players know how to adjust.

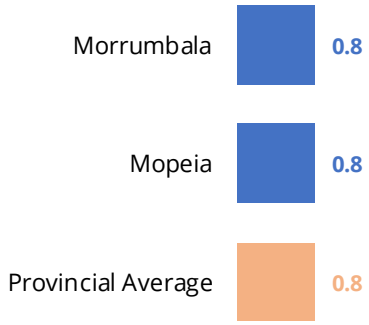
KEY FINDINGS

THE AVERAGE PROVINCIAL SCORE FOR PRICE WAS 0.8 (out of a maximum of 10), which indicates a VERY HIGH RISK for cash-based interventions in the assessed markets. The markets presented serious price stability issues (for all the three (3) commodity categories) as traders feel they cannot meaningfully predict the prices of items they are selling for the following week, indicating some volatility and resulting in an index considerably below acceptable levels.

Significant affordability issues for the main food category - cereal - has been observed as 63% of the traders reported cereal price increase, mainly rice (66%) and cassava (30%), while 59% of the traders reported an increase in prices of food other than cereals. In this case concerning food groups are Oil & Fats and herbs, condiments and spices as 31% of traders reported price increase. Regarding non-food items: 32 % of the traders reported rising non-food prices (WASH, communication, Education and household items).

The vast majority of retailers reported important issues with price stability as they are not able to predict short and long-term price variations for any of the commodities assessed. In fact, 81% of retailers are not able to predict cereal prices, 73% for food other than cereals and 74% for non-food items.

SCORE BY MARKET



FINDINGS BY MFI DIMENSION : PRICE (cont'd)

Figure 5.



FINDINGS BY MFI DIMENSION : PRICE (cont'd)

Figure 6.



FINDINGS BY MFI DIMENSION : SUPPLY CHAIN RESILIENCE

SUPPLY CHAIN RESILIENCE

The resilience dimension evaluates responsiveness of supply chains and their resilience or vulnerability to shock. The score addresses two questions:

1. How responsive are supply chains to shock in demand?
2. How vulnerable is the Supply Chain?

Responsiveness of supply chains is measured by stock and lead time. A shorter lead time helps a trader adjust to changing demand more rapidly and to be aware of upstream supply chain disruptions earlier. Additionally, a shorter lead time is associated with lower stocks.

The MFI, thus, takes a separate look at buffer stocks, rating their level as adequate or not (i.e. sufficient for 1 week) and ability to restock rapidly.

The MFI also evaluates the suppliers' traders rely on. The score considers location and number of supply sources. Supply chains are more resilient with more supply sources and geographically dispersed suppliers.

KEY FINDINGS

THE AVERAGE PROVINCIAL SCORE FOR SUPPLY CHAIN RESILIENCE REACHED A MODEST 5.8, reflecting the barely acceptable resilience of supply chains and that markets have moderate ability to respond to operational disruptions through flexible contingency planning and forecasting.

- Both districts in Zambezia presented issues related to node density, showing that retailers have suppliers in near locations and therefore are vulnerable to possible shocks, regarding node complexity Mopeia shows that most traders are currently purchasing from one supplier for cereal foods and NFI and in Morrumbala the same was reported for NFI.

- Markets scored very high on supply chain responsiveness, indicating that they are able to respond to rapid change in demand.
  - Most of assessed retailers have stocks that last more than a week
  - They are able restock or to cope with an increase in demand in a reasonable time (generally orders are filled within a week).

SCORE BY MARKET



FINDINGS BY MFI DIMENSION : SUPPLY CHAIN RESILIENCE

Table 2.

	Node complexity [1]			Node criticality [2]			Node Density [3]			Responsiveness	
Market Name	Cereal Food	Non-Cereal Food	NFI	Cereal Food	Non-Cereal Food	NFI	Cereal Food	Non-Cereal Food	NFI	Current stock	Lead time
Mopeia	No	Yes	No	No	No	No	Yes	Yes	Yes	Yes	Yes
Morrumbala	Yes	Yes	No	Yes	No	No	Yes	Yes	Yes	Yes	Yes

Node:

(1) Node Complexity - both markets presented issues related to traders using a single suppliers for NFI, in Mopeia also for Cereal food traders mostly have a single supplier.

(2) Node Criticality - Morrumbala reported presence of a single supplier on which most traders rely upon for cereal food.

(3) Node Density - geographic distribution of suppliers was found to be an issue in the supply chain network for all the three (3) categories – cereals food, non-cereal food and NFI – in all the assessed markets in Zambezia (Mopeia and Morrumbala).

Responsiveness:

Supply chain responsiveness is very critical to meet customer demands effectively and all the assessed markets reported being capable to restock with a short Lead Time (one week period) - which encompasses the entire duration from order placement to delivery in a shop - preventing not only WFP (potential) beneficiaries and the entire range of clients from the long-term waiting but building also a more efficient supply chain that is better equipped to meet the demands of today's fast-paced market associated to the fact that all the traders reported enough Current Stock to last at least a week, considering their customers' current demand.

Note:

[1]The question verbatim for Node Complexity reads: *"D2.2 Do you currently have more than 1 supplier for: [Cereal food, other food, and non-food items]?"*

[2]The question verbatim for Node Criticality reads: *"D2.3 Does your business mostly rely upon a single supplier for: [Cereal food, other food, and non-food items]?"*

[3]The question verbatim for Node Density reads: *"D2.1 Are most of your suppliers geographically located in the same place for: [Cereal food, other food, and non-food items]?"*

FINDINGS BY MFI DIMENSION : COMPETITION

COMPETITION

Competition among market participants characterizes a functioning market. This dimension investigates if the business environment is conducive to fair competition, which is typically the case when there are enough traders in the marketplace with equal to varying power.

The Competition score addresses two questions:

1.Is there sufficient competition within the market? *Are there more than 5 players?*

2.Does a trader control the market? *Is there a monopoly?*

These two conditions should lower the risk for few traders to build a monopoly, set prices and profit disproportionally from a potential introduction of cash-based transfers

KEY FINDINGS

THE AVERAGE PROVINCIAL SCORE FOR COMPETITION IS 7.5, reflecting a good level of competition in the assessed markets and the ability of the market to anticipate customers' needs and preferences.

COMPETITION - KEY FINDINGS:

i) Markets in Mopeia present issues with competition only for Food Other Than Cereal, for which there are less than 5 retailers in the markets and usually one retailer controlling the market.

ii) In Morrumbala it was found instead that there is low competition for Non-food Items, for which there are less than 5 retailers with 1 controlling the market

SCORE BY MARKET



FINDINGS BY MFI DIMENSION : COMPETITION (cont'd)

Table 4.

	Less than five retailers in the market			One retailer that controls the market		
Market Name	**Cereal Food**	**Non-Cereal Food**	**NFI**	**Cereal Food**	**Non-Cereal Food**	**NFI**
Mopeia	No	Yes	No	No	Yes	No
Morrumbala	No	No	Yes	No	No	Yes

FINDINGS BY MFI DIMENSION : INFRASTRUCTURE

INFRASTRUCTURE

The infrastructure dimension assesses the type and condition of the physical structures which host shops in addition to sewage systems, electricity and communication networks in the market premises. Infrastructure addresses:

1. What is the overall condition of the shops where customers shop?

2. What infrastructure is available in and around the shops?

For question 1, the methodology scores the market based on the state of the shops. population evaluated according to three categories: Poor state and severe maintenance issues, medium state (minor maintenance issues) and Good state.

For question 2, the methodology verifies the presence of the following services in the market premises:

- A closed sewage system
- A dedicated waste collection area
- Walkways and emergency exit(s)
- Shelter
- Uninterrupted electricity
- Reliable communication network (mobile phone coverage and/or internet)
- Water availability (manual pump or tap in close proximity of each vendor, i.e. max 5 min walking)
- Toilets

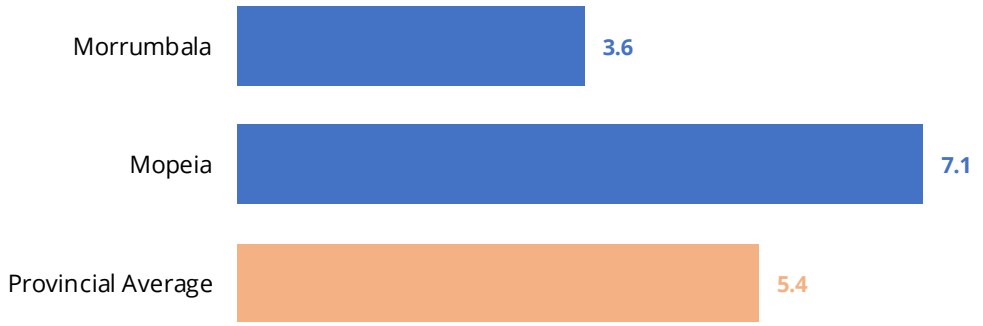
KEY FINDINGS

THE AVERAGE PROVINCIAL SCORE FOR INFRASTRUCTURE is 5.4 - as most of the traders operate from semi-permanent structures, with the majority in medium conditions requiring several maintenance issues. Overall markets in Morrumbala presents much lower infrastructure that those assessed in Mopeia and require important adjustments.

The most preoccupying factor is the fact that most of the services expected in a market premise are missing:

- None of the market has closed sewage systems;
- None of the market has walkways;
- None of the market has a dedicated area for waste collection;
- Morrumbala reported no water available in or in the proximity of the marketplace and no toilets

SCORE BY MARKET



FINDINGS BY MFI DIMENSION : INFRASTRUCTURE

Table 5.

	Conditions [1]			Features [2]						
Market Name	Good	Medium	Poor	Closed sewage system	Network	Shelter	Toilet	Wakways	Waste Collection Area	Water
Mopeia	Yes	No	No	No	Yes	No	Yes	No	No	Yes
Morrumbala	No	Yes	No	No	Yes	Yes	No	No	No	No

- Mopeia were found with structures in good conditions, all the others presented semi-permanent structures poorly maintained.
- Reliable communication network was found in the markets assessed.
- Important gaps in sewage systems, walkways and waste collection were found in both markets.
- Lack of proper shelters for customers and other market players in Mopeia.
- Lack of Toilets and water pumps in Morrumbala in the close proximity were also noticed, which requires important interventions before implementing Market-Based Interventions.

Note:

[1] **Conditions:** The combination of "Yes", "No" attempts to give a description of the state of traders. As an example, the combination “Yes“ ,"No", “No" (Order matters!) can be interpreted as the best one as it depicts a market where all stores are in an excellent condition.

[2] **Features**: The table displays a “Yes” if the considered features are present in a market, a “No” otherwise.

FINDINGS BY MFI DIMENSION : SERVICE

SERVICE

The service dimension looks at the service provided while shopping as well as during check-out. The level of service is directly or indirectly associated with different features of a functioning market such as transparency, competition or reliability.
The more competitive the market environment, the higher incentive for retailers to provide the best value to customers; otherwise, these might shop somewhere else.

The service dimension addresses:

1. How is the customers' overall shopping experience?
2. How is the check-out experience?

The methodology includes in shopping experience the following elements: overall shop organization – which influences how easy it is to find products; price transparency – tags; possibility to shop remotely. Similarly, we consider costumers check-out experience based on the following elements: accepted forms of payments, waiting time at check out and itemized receipts.

KEY FINDINGS

THE AVERAGE PROVINCIAL SCORE FOR SERVICE is 3.3. all the markets in all assessed areas of Zambezia have similar characteristics in terms of services provided during shopping and at the check-out time. According to this findings markets characteristics are not very conducive for CBT activities because they present important challenges that could be exacerbated by the increased demand.

Service - key findings:

i) In terms of shopping experience, traders in both markets have products displayed for easy identification and selection.

Ii) They do not provide the possibility to shop remotely

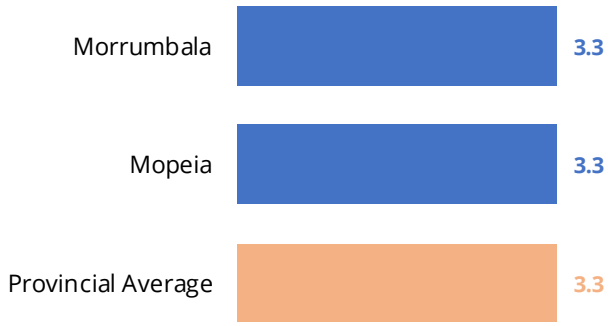
Iii) they do not have visible price tags

iv) Most of the assessed retailers didn't have automatic receipt devices, which could increase considerably waiting time if the demand increases suddenly.

v) They accept other forms of payment than cash

vi) Usually, considerable waiting time is observed at check out

SCORE BY MARKET



FINDINGS BY MFI DIMENSION : SERVICE (cont'd)

Table 6.

	Shopping Service			Checkout Service		
Market Name	**Product display**	**Remote purchase**	**Visible price tags**	**Automatic receipt issuance**	**Multiple payment Forms**	**Waiting time**
Mopeia	Yes	No	No	No	Yes	No
Morrumbala	Yes	No	No	No	Yes	No

FINDINGS BY MFI DIMENSION : FOOD QUALITY

FOOD QUALITY

In many marketplaces, food may be available in sufficient quantities, but the quality of this food may vary. Availability of food unsafe for human consumption is no better – even worse, in some instances – than unavailability of food. The MFI examines food quality against key principles of hygiene and cleanliness, material separation, temperature control and stock management. Essentially, the survey examines 8 key conditions for shops:

1. Is food protected from contaminants?
2. Are fresh fruits and vegetables in the shops well-separated from raw meat, poultry, fish or seafood?
3. Is food that requires refrigeration properly refrigerated?
4. Is refrigeration in the shops always working?
5. Is food for sale in the shop not expired?
6. Are processed pre-packaged foods in the shops intact and in properly labelled containers?
7. Is food in the shops free of visible signs of spoilage and bad smells?
8. Are food packages intact and free from signs of decay or damage?

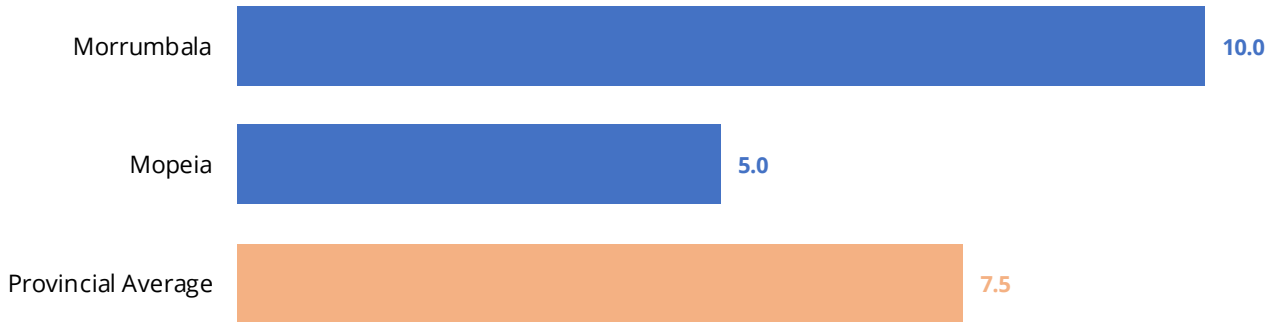
KEY FINDINGS

THE AVERAGE PROVINCIAL SCORE FOR FOOD QUALITY WAS 7.5, reflecting an acceptable level of food quality in the assessed markets. When analyzing separately results in the two districts, it's clear that Mopeia presents important issue related to food quality not found in Morrumbala, which scored perfectly in this dimension.

In Mopeia the following issues were found in most of the assessed retailers:

- Products not presenting labels with expiration date and, in some cases, products were found expired
- Fresh fruits and vegetables in the shops are not well-separated from raw meat, poultry, fish
- Some products requiring refrigeration, such as meat and poultry are not always refrigerated despite the fact that most shops has working refrigerators.
- Processed pre-packaged foods in the shops are not always intact and in properly labelled containers.

SCORE BY MARKET



FINDINGS BY MFI DIMENSION : FOOD QUALITY (cont'd)

Table 4.

Market Name	All Goods are labelled and not exceeding the expiry date	Food has no visible sign of Spoilage	Food separated from contaminants	Fresh separated from animal origin	Meats/Poultry Food always refrigated	Plastic packaged food always intact	Prepackaged intact&labelled	Refrigeraion always working	% Essential Food Quality Features
Mopeia	No	Yes	Yes	No	No	Yes	No	Yes	50%
Morrumbala	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	100%

Note: The table above displays a "Yes" if the analysed feature of Food quality is predominantly present in a market, a "No" otherwise. Considering the results, find the % Essential Food quality Features to summarize the information.

FINDINGS BY MFI DIMENSION : ACCESS & PROTECTION

ACCESS & PROTECTION

Safe access to the marketplace is a crucial dimension of market functionality. Access addresses two key questions:

1. Are beneficiaries and customers able to reach markets?
2. Are there fragile groups facing threat and protection issues accessing the market? Or are there any general security concerns?

The MFI weights Access and Protection issues by severity. For access it weights based on the level of severity: is it hard to get to the market because of road infrastructure, or is access limited due to seasonal events? Were there natural disasters preventing costumers access more permanently? The weight is higher the more permanent the access issues are.

The protection rationale is similar: issues are weighted according to the fraction of the population impacted: Are there social barriers? Are there protection issues limited to a marginalized group?

KEY FINDINGS

THE ACCESS AND PROTECTION DIMENSION SCORED ON AVERAGE VERY HIGH at 8.8 showing that this dimension doesn't pose risks for the functionality of the markets. In both districts market access doesn't seem particularly influenced by the seasonality and impact of natural disasters, there are good security conditions and no social barriers preventing access to the markets, Nevertheless, a couple of issues requiring attention were found:

- In Mopeia markets were found quite distant from the main roads
- In Morrumbala some markets presented physical threats causing difficulties in accessing the markets for disabled people

SCORE BY MARKET



RECOMMENDATIONS/ADVISED MITIGATION

MOPEIA DISTRICT

- Low-performing price dimension
    1. Strengthen price monitoring activities to ensure regular update of price data;
    2. Increase the frequency of distribution by dividing the beneficiaries into groups receiving at different times during the month;
    3. Vouchers with capped prices agreed with traders,

- Lack of competition asks for vouchers with price control and introducing new traders when possible

- Weak resilience
    1. Preparation of a contingency plan based on in-kind,
    2. Secure predictable demand for contracted traders to allow them to stock in advance

- Weak Services - Set up Retailer Capacity Strengthening Training as part of the Market Development Activities (MDA)

- Quality assurance
    1. Contracting traders that operate with adequate food safety standards
    2. Mobilizing third parties/local authorities to build retailer capacity and improve compliance with local food safety regulations
    3. Offer food safety/quality certification to targeted retailers as capacity building activity

RECOMMENDATIONS/ADVISED MITIGATION

MORRUMBALA DISTRICT

- Low-performing price dimension
    1. Strengthen price monitoring activities to ensure regular update of price data;
    2. Increase the frequency of distribution by dividing the beneficiaries into groups receiving at different times during the month;
    3. Vouchers with capped prices agreed with traders,

- Weak resilience
    1. Preparation of a contingency plan based on in-kind,
    2. Secure predictable demand for contracted traders to allow them to stock in advance

- Low infrastructure
    1. Contracting traders operating in premises with adequate standards,
    2. Mobilising stakeholders to make required investments,
    3. WFP to build/rehabilitate market infrastructure (RIAB),
    4. Support smaller retailers to get the licenses, thus, ensuring food quality is up to the standards

- Weak Services - Set up Retailer Capacity Strengthening Training as part of the Market Development Activities (MDA)

CONTACT INFORMATION

END!

Research, Analysis and Monitoring (RAM) team – WFP Mozambique:

Francesco Stompanato francesco.stompanato@wfp.org

Panfilio Malamule panfilio.malamule@wfp.org

